A rational contour proof of a Ramanujan-type $1/\pi^2$ identity

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Abstract

We evaluate an absolutely convergent weighted factorial series by reducing its loaded radial integral to the square of a native Gauss value. A rational coefficient one-form provides a contour flux; a fixed rational correction turns that flux into an antiderivative of the entire integrand. We justify its removable interior singularity, prove that the upper endpoint vanishes, and compute the lower endpoint as a full positive residue. A signed algebraic Gauss transformation and an absolutely normalized Wronskian then supply the required $1/\pi$ value. All rational inputs, the ninety-coefficient witness, and a complete executable certificate are included in this standalone manuscript.

Claim and verification boundary. The claim is the literal scalar equality in Theorem 1.1, including absolute convergence. Its retained human-checkable derivation is Complete, with no open mathematical premise in that scalar scope. Evidence is [ARG] for the analytic argument and [ONE] for the retained exact rational checks. The 7 October 2026 document review found no substantive defect in its reviewed scope. Human mathematical acceptance remains Pending. The same scalar equality has a completed Lean formalization as of 7 October 2026, including unconditional convergence and sum statements. These statements concern the scalar identity only.

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1 The statement and the proof structure

Theorem 1.1. *The series on the left converges absolutely, and*

$$R := \sum_{n=0}^{\infty} \frac{(6n)!}{(n!)^6} \frac{532n^2 + 126n + 9}{10^{6n}} = \frac{375}{4\pi^2}. \tag{1}$$

The central reduction is

$$R = \frac{3}{4}\{3F(u) + 28\theta F(u)\}^2, \qquad u = \frac{27}{125}, \tag{2}$$

where the native function and its Euler derivative are

$$F(q) = \sum_{n\geq 0} \frac{(1/6)_n (1/2)_n (5/6)_n}{(n!)^3} q^n, \qquad \theta F(q) = q\frac{\mathrm{d}F}{\mathrm{d}q}(q).$$

Here $(a)_0 = 1$ and $(a)_n = a(a+1)\cdots(a+n-1)$ for $n \geq 1$. The separate, absolutely normalized Gauss calculation gives

$$3F(u) + 28\theta F(u) = \frac{5\sqrt{5}}{\pi}. \tag{3}$$

Thus the $1/\pi^2$ value comes from the square of an independently evaluated $1/\pi$ expression. We prove both equalities below.

Set

$$\rho = \frac{729}{15625} = \frac{6^6}{10^6}, \qquad u = \frac{27}{125} = \sqrt{\rho}, \qquad \tau = 2\pi i, \tag{4}$$

and distinguish the original series weight from the radial load:

$$P(z) = 9 + 126z + 532z^2, \qquad Q(z) = (1+6z)P(z) = 9 + 180z + 1288z^2 + 3192z^3. \tag{5}$$

The radial moment replaces the factorial coefficients by the squared coefficients of F and introduces the factor $1 + 6z$. Let a_n denote the coefficient of q^n in F, and put $h(\lambda) = \sum_{n\geq 0} a_n^2 \lambda^n$ and $\theta_\lambda = \lambda\,\mathrm{d}/\mathrm{d}\lambda$. The whole series becomes

$$R = \int_0^1 Q(\theta_\lambda) h\big(\rho(1-U^2)^3\big)\,\mathrm{d}U.$$

In this notation the Euler operator acts on h before evaluation at the displayed radial argument.

To evaluate this integral, we construct an actual function $\mathcal{E}$ with

$$\mathcal{E}'(U) = -\frac{1}{3375} Q(\theta_\lambda) h(\lambda), \qquad \lambda = \rho(1-U^2)^3.$$

Its contour flux initially has a finite-coordinate derivative remainder. An explicit rational correction removes that remainder. We then show that the apparent interior pole is removable and that

$$\mathcal{E}(1-) = 0, \qquad \mathcal{E}(0+) = \frac{\{3F(u) + 28\theta F(u)\}^2}{4500}.$$

The fundamental theorem of calculus gives (2), because $3375/4500 = 3/4$. The rational identities alone would not justify this step: the contour deformation, escape, and endpoint limits are essential parts of the proof.

Step	Mathematical task	Location
Radial and angular extraction	Identify the entire factorial series with the loaded Hadamard integral, including $Q = (1+6z)P$.	Section 2
Coefficient form	Construct a flux with separated moving poles and a covariant source in the native product system.	Section 3
Two reductions	Remove an angular total derivative, then correct the radial remainder to obtain $\mathcal{E}'$.	Section 4; Appendix A
Boundary analysis	Remove the interior escape, prove the zero upper limit, and compute the full lower residue.	Sections 5–7
Gauss normalization	Prove the signed transformation and fix the absolute Wronskian scale in (3).	Section 8
Assembly	Apply the endpoint formula and identify the original series.	Section 9

2 From the literal series to the loaded integral

For $n \geq 0$ define

$$a_n = \frac{(1/6)_n (1/2)_n (5/6)_n}{(n!)^3} = \frac{(6n)!}{(3n)!(n!)^3\, 1728^n}, \tag{6}$$

where $(a)_0 = 1$ and $(a)_n = a(a+1)\cdots(a+n-1)$. The equality of the two formulas follows from their common value 1 at $n = 0$ and their common successive ratio

$$\frac{a_{n+1}}{a_n} = \frac{(n+1/6)(n+1/2)(n+5/6)}{(n+1)^3}. \tag{7}$$

In particular $0 < a_n \leq 1$. Define, on the complex unit disk,

$$F(q) = \sum_{n\geq 0} a_n q^n, \qquad h(\lambda) = \sum_{n\geq 0} a_n^2 \lambda^n, \qquad \theta = q\frac{\mathrm{d}}{\mathrm{d}q}. \tag{8}$$

The variable attached to each Euler operator will always be stated. All finite Euler derivatives converge normally on compact subsets of their unit disks, by the majorant $\sum n^j r^n$ for $r < 1$.

Lemma 2.1 (Radial moments)**.** *For every integer $m \geq 0$,*

$$I_m := \int_0^1 (1-U^2)^m \,\mathrm{d}U = \frac{4^m (m!)^2}{(2m+1)!}.$$

Consequently

$$(6n+1) a_n^2 I_{3n} = \frac{(6n)!}{6^{6n}(n!)^6}. \tag{9}$$

Proof. The derivative of $U(1-U^2)^{m+1}$ has integral zero: the function vanishes at both endpoints. Its integral gives $(2m+3)I_{m+1} = (2m+2)I_m$. Starting at $I_0 = 1$ proves the stated formula. Substitution of (6) with $m = 3n$ proves (9); here $4^{3n}/1728^{2n} = 6^{-6n}$. □

Absolute convergence of the original series follows independently from $(6n)!/(n!)^6 \le 6^{6n}$, since this multinomial coefficient is one term in the expansion of $(1+\cdots+1)^{6n}$ with six summands. Thus its terms are bounded by $P(n)\rho^n$, a summable sequence. Put $x = 1-U^2$ and $\lambda = \rho x^3$. For $0 \le U \le 1$, the loaded summands are bounded by $Q(n)\rho^n$. This justifies termwise radial integration and yields the exact identity

$$R = \int_0^1 Q(\theta_\lambda)h(\rho(1-U^2)^3)\,\mathrm{d}U. \tag{10}$$

Indeed the nth integrated term is $Q(n)a_n^2\rho^n I_{3n}$, which equals the nth original term by (9). The term $n = 0$ has value 9 in both expressions.

Lemma 2.2 (Angular coefficient extraction)**.** *For $0 < \lambda < 1$ and $\lambda < r < 1$, every positively oriented circle $|s| = r$ satisfies*

$$\theta_\lambda^j h(\lambda) = \frac{1}{\tau}\oint_{|s|=r} F(s)(\theta_b^j F)(\lambda/s)\,\frac{\mathrm{d}s}{s}, \qquad j \ge 0. \tag{11}$$

Here $\theta_b = b\,\mathrm{d}/\mathrm{d}b$, and for $j = 0$ the factor is $F(b)$.

Proof. Both native power series and their required Euler derivatives converge absolutely and uniformly on the circle. Expanding the product, the normalized integral of $s^{m-n}\mathrm{d}s/s$ equals 1 when $m = n$ and 0 otherwise. This leaves $\sum a_n^2 n^j \lambda^n$. For $j = 0$, the factor n^0 denotes 1, including at $n = 0$. The bounds already given also justify local parameter differentiation. □

3 The rational coefficient form and its contour flux

The purpose of the coefficient form is to produce a contour flux whose radial derivative differs from the target load by a rationally correctable remainder. The flux by itself is not yet the required antiderivative.

3.1 Native, ordinary symmetric-square, and dual frames

The coefficient recurrence gives the native differential equation

$$\big[\theta^3 - q(\theta+1/6)(\theta+1/2)(\theta+5/6)\big]F = 0.$$

The native frame records the Euler jet $Y(q) = (F(q), \theta F(q), \theta^2 F(q))^{\mathsf{T}}$. Its equation is

$$\theta Y = N(q)Y, \qquad N(q) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ \dfrac{5q}{72(1-q)} & \dfrac{23q}{36(1-q)} & \dfrac{3q}{2(1-q)} \end{pmatrix}. \tag{12}$$

An ordinary symmetric-square frame has coordinates $(y^2, 2yy', y'^2)^{\mathsf{T}}$, where the primes are ordinary q derivatives and $y'' = -a(q)y' + c(q)y$. Its connection is

$$\mathsf{A}(q) = \begin{pmatrix} 0 & 1 & 0 \\ 2c(q) & -a(q) & 2 \\ 0 & c(q) & -2a(q) \end{pmatrix}, \quad a(q) = \frac{1-3q/2}{q(1-q)}, \quad c(q) = \frac{5}{144q(1-q)}. \tag{13}$$

For example, differentiating $2yy'$ gives $2c\,y^2 - a(2yy') + 2y'^2$, which is the second row. Section 8 later proves $F = g^2$ for the normalized native Gauss solution. No unproved symmetric-square normalization is used here.

The explicit bridge to the dual ordinary frame is

$$T(q) = \begin{pmatrix} -5q/72 & 0 & 2q^2(1-q) \\ -q/2 & q(q-1) & 0 \\ 1-q & 0 & 0 \end{pmatrix}, \qquad W(q) = T(q)^{\mathsf{T}} Y(q). \tag{14}$$

The rational identity $T' + N^{\mathsf{T}} T/q + T\mathsf{A} = 0$ implies

$$W'(q) = -\mathsf{A}(q)^{\mathsf{T}} W(q). \tag{15}$$

It fixes the transpose and sign conventions for pairing a matrix coefficient with $W(s)$ and $W(\lambda/s)$. Appendix A also gives the ordinary-to-Euler gauge explicitly.

3.2 Moving poles and the complete one-form

For $0 < U < 1$ set

$$\begin{aligned} &x = 1 - U^2, && \lambda = \rho x^3, && b = \lambda/s, \\ &m = \frac{27x(80 - 99x + 27x^2)}{1000}, && \ell_0 = \frac{27x(9x-16)}{1000}, && \ell = U\ell_0, \\ &M = 25 - 9x, && d = x^4 M(99x-50)^2, && f = s^2 - 2ms + \lambda. \end{aligned} \tag{16}$$

The identity $m^2 - \lambda = \ell^2 M$ factors f into two moving poles:

$$\sigma_\pm = m \pm \ell\sqrt{M}, \qquad 0 < \sigma_+ < \sqrt{\lambda} < \sigma_-. \tag{17}$$

Here m is their midpoint, $\ell\sqrt{M}$ is their signed half-separation, and $\lambda = \sigma_+\sigma_-$ is their product. On $0 < x < 1$, $m > 0$, $\ell_0 < 0$, and $M > 0$, so the label $+$ denotes the smaller root. Positivity follows from their positive sum and product; their distinctness and product give the displayed separation. The balanced circle $|s| = \sqrt{\lambda}$ therefore encloses exactly σ_+. On it both native arguments s and b have modulus $\sqrt{\lambda} < 1$, which is also useful for endpoint estimates.

The fixed polynomial matrices $L_P(x), R_P(x)$ are printed in Appendix A. They are retained source-lift data; their role in this proof is determined by the exact identities checked below, rather than by a claimed coefficient-discovery procedure. Define the complete matrix-valued one-form $\beta_N = B_s\,\mathrm{d}s + B_U\,\mathrm{d}U$ by

$$B_s(U,s) = \frac{2UL_P(s-m) + 2\ell M R_P}{df}, \tag{18}$$

$$B_U(U,s) = \frac{UL_P\,\partial_U f - 2R_P\{\ell M \partial_U m + (s-m)(M\partial_U \ell + 9U\ell)\}}{df}. \tag{19}$$

The partial U derivatives hold s fixed. Both components and both summands of B_s are needed.

The residues become visible through the partial fractions

$$B_s = \frac{UL_P + \sqrt{M}R_P}{d(s-\sigma_+)} + \frac{UL_P - \sqrt{M}R_P}{d(s-\sigma_-)}. \tag{20}$$

Indeed the numerator over df is $UL_P(2s - \sigma_+ - \sigma_-) + \sqrt{M}R_P(\sigma_+ - \sigma_-)$, which is exactly $2UL_P(s-m) + 2\ell M R_P$. There is also a useful local differential description:

$$\beta_N = \frac{UL_P}{d}\,\mathrm{d}\log f + \frac{\sqrt{M}R_P}{d}\,\mathrm{d}\log\frac{s-\sigma_+}{s-\sigma_-}. \tag{21}$$

Here $\mathrm{d}\log g$ means $\mathrm{d}g/g$ on the regular chart; no global logarithm is chosen. If $z = s - m$ and $\delta = \ell\sqrt{M}$, then

$$\mathrm{d}\log\frac{z-\delta}{z+\delta} = \frac{2\delta\,\mathrm{d}z - 2z\,\mathrm{d}\delta}{f}.$$

Using $\mathrm{d}M = 18U\,\mathrm{d}U$ expands (21) into both (18) and (19). In particular, differentiating $\sqrt{M}$ supplies the $9U\ell$ term in B_U.

The actual scalar flux is the normalized, positively oriented integral

$$\mathfrak{b}(U) = \frac{1}{\tau}\oint_{|s|=\sqrt{\lambda}} W(s)^{\mathsf{T}} B_s(U,s) W(\lambda/s)\,\mathrm{d}s. \tag{22}$$

Its coefficients also have a rational escape pole at

$$99x - 50 = 0, \qquad U_e = \frac{7}{3\sqrt{11}}. \tag{23}$$

For now the flux is defined off this point. At the lower endpoint the moving poles coalesce at u. Their residue matrices tend to $\pm 4R_P(1)/d(1)$, while B_s tends to zero at each fixed $s \neq u$. The convergence is not uniform on the shrinking gap around the poles; Section 7 computes the surviving full residue by contour deformation.

3.3 The covariant source has no moving f-poles

Radial differentiation acts on both the coefficient and the second dual solution. Angular differentiation acts on both dual solutions. By (15) the required coefficient derivatives are

$$\mathcal{D}_U B_s = \partial_U B_s + \frac{6U}{x} B_s\, b\mathsf{A}(b)^{\mathsf{T}},$$
$$\mathcal{D}_s B_U = \partial_s B_U - \mathsf{A}(s) B_U + \frac{b}{s} B_U \mathsf{A}(b)^{\mathsf{T}}.$$

The chain-rule factors are $\partial_U b = -6Ub/x$ and $\partial_s b = -b/s$. The complete native source matrix is

$$\mathsf{S} = T(s)(-\mathcal{D}_U B_s + \mathcal{D}_s B_U) T(b)^{\mathsf{T}}. \tag{24}$$

The B_U component supplies the angular total derivative in this formula. Although the form has moving f-poles, its complete covariant source cancels them. The certificate clears the fixed connection denominators in $-\mathcal{D}_U B_s + \mathcal{D}_s B_U$ and divides each of the nine resulting numerators by f^2, with zero remainder. The T matrices introduce no new moving f-poles, so the native source row has angular poles only at $0, 1, \lambda, \infty$. This checked cancellation is the necessary input to the angular elimination in the next section; it is not assumed from the pole shape of the form.

4 Angular reduction and the radial correction

There are two exactness steps. First, integration by parts converts the complete source to six specific contour functions. Second, a rational row corrects their radial contribution to the desired load.

4.1 Integration by parts and the chosen representatives

Let $J = N(0)$, $\mathbf{w} = (5/72, 23/36, 3/2)$, and $E = \mathbf{e}_2^{\mathsf{T}}\mathbf{w}$, where $\mathbf{e}_0, \mathbf{e}_1, \mathbf{e}_2$ are the standard rows of length three. With I the 3 by 3 identity, the native product column $Z = Y(s) \otimes Y(\lambda/s)$ satisfies

$$Z_s = \mathcal{M}_s Z, \qquad \mathcal{M}_s = \frac{N(s)\otimes I - I \otimes N(\lambda/s)}{s}.$$

Consequently every rational coefficient row η satisfies

$$(\partial_s \eta + \eta \mathcal{M}_s) Z = \frac{\mathrm{d}}{\mathrm{d}s}(\eta Z). \tag{25}$$

Its contour integral is zero whenever the pairing is single-valued and regular on the circle. Regularity throughout the interior is not required. This is the integration-by-parts identity used below. We flatten 3 by 3 matrices row by row, in tensor order $(00, 01, 02, 10, 11, 12, 20, 21, 22)$.

The connection has residues $R_1 = -E \otimes I$ at 1 and $R_\lambda = -I \otimes E$ at λ. Each has a three-dimensional row image. Their tensor shapes motivate the two triples

$$\Psi_j = \frac{\mathbf{w} \otimes \mathbf{e}_j}{s-1}, \qquad \Psi_{j+3} = \frac{\mathbf{e}_j \otimes \mathbf{w}}{s-\lambda}, \qquad 0 \le j \le 2. \tag{26}$$

Define the actual contour functions

$$v_j(\lambda) = \frac{1}{\tau} \oint_{|s|=r} \Psi_j(s,\lambda) Z(s,\lambda)\, \mathrm{d}s, \qquad 0 < \lambda < r < 1, \quad \mathbf{v} = (v_0, \dots, v_5)^{\mathsf{T}}. \tag{27}$$

Their integrands are analytic in the annulus between any two admissible circles, so the integrals are independent of r. For complex λ, choose $|\lambda| < r < 1$; a fixed-circle argument gives local holomorphy on the punctured unit disk.

4.2 Anchor and parameter action fix the actual state

Define the covariant parameter action on rows by

$$D_\lambda \psi = \lambda \partial_\lambda \psi + \psi(I \otimes N(\lambda/s)).$$

It corresponds to differentiating their pairings by θ_λ. Appendix A gives the explicit constant matrix C, with rows $c_0, \dots, c_5$, and the rational matrix B_6. The anchor is a rational identity

$$\frac{e_{00}}{s} = \sum_{j=0}^{5} (c_0)_j \Psi_j + \partial_s \eta_0 + \eta_0 \mathcal{M}_s, \qquad \eta_0 = e_{00} R_0^{-1},$$

where $e_{00} = \mathbf{e}_0 \otimes \mathbf{e}_0$ and R_0 is the residue at 0. Its normalized contour integral is $h = c_0 \mathbf{v}$, by the angular extraction in Lemma 2.2 and (25). The six parameter identities have the form

$$D_\lambda \Psi_j = \sum_{k=0}^{5} (B_6)_{jk} \Psi_k + \partial_s p_j + p_j \mathcal{M}_s.$$

Their explicit primitives are printed in the appendix. Integration therefore gives the actual analytic action

$$\theta_\lambda \mathbf{v} = B_6(\lambda) \mathbf{v}. \tag{28}$$

A worked row makes the denominators and block signs concrete. Since $\mathbf{e}_2 J = 0$ and $\mathbf{e}_2 E = \mathbf{w}$,

$$\begin{aligned} D_\lambda \Psi_2 &= \frac{\lambda\, \mathbf{w} \otimes \mathbf{w}}{(s-1)(s-\lambda)} \\ &= \frac{\lambda}{1-\lambda} \left(\frac{\mathbf{w} \otimes \mathbf{w}}{s-1} - \frac{\mathbf{w} \otimes \mathbf{w}}{s-\lambda} \right). \end{aligned}$$

This is row 2 of B_6 (indices start at zero); its angular primitive is zero. The other rows are checked by the same explicit row identities. Finally $c_j B_6 = c_{j+1}$ for $0 \le j < 5$ and $\det C = 373248/125 \neq 0$. Repeated differentiation of the anchored identity gives

$$C\mathbf{v} = (h, \theta_\lambda h, \dots, \theta_\lambda^5 h)^{\mathsf{T}}. \tag{29}$$

Thus C identifies the six specified contour functions with the normalized series jet. It fixes their analytic scale rather than assuming an unspecified period normalization.

4.3 Reduction of the complete source

The pole cancellation in Section 3 permits deterministic angular elimination on flatten(S). The appendix describes the steps: subtract covariant derivatives to remove higher finite poles, remove the polynomial part at infinity, and remove the simple residue at zero using invertible R_0. All required inverses exist by the printed residue spectra, and each step lowers a pole order or polynomial degree.

For an arbitrary rational input row with poles only at these four punctures, this procedure leaves $r_1/(s-1) + r_\lambda/(s-\lambda)$, with two arbitrary length-nine residues: an eighteen-coordinate normal form. For the present complete source the certificate additionally proves

$$r_1 = \mathbf{w} \otimes (G_0, G_1, G_2), \qquad r_\lambda = (G_3, G_4, G_5) \otimes \mathbf{w}.$$

These factorizations place the two residues in three-dimensional tensor subspaces. They are substantive identities of this source and do not assert a generic six-dimensional reduction. With the rational primitive η constructed by the elimination, the full result is

$$\text{flatten}(\mathsf{S}) = \sum_{j=0}^{5} G_j(x)\Psi_j + \partial_s \eta + \eta \mathcal{M}_s. \tag{30}$$

The certificate checks every entry, retains the primitive, and verifies both tensor factorizations. Its coefficient-field scaling is undone before G and η are used.

For $U_0 \in (0,1) \setminus \{U_e\}$, root separation allows a fixed radius near U_0 without crossing an f-pole. The integrands and their parameter derivatives are analytic on this compact circle. By the dual equation,

$$\mathfrak{b}'(U) = \frac{1}{\tau} \oint W(s)^{\mathsf{T}} (\mathcal{D}_U B_s) W(b)\, \mathrm{d}s.$$

The integral with $\mathcal{D}_s B_U$ is zero: its integrand is the s derivative of $W(s)^{\mathsf{T}} B_U W(b)$. Subtracting this derivative and using the sign in (24), then (30) and (25), gives

$$\mathfrak{b}'(U) = -G(x)\mathbf{v}(\lambda). \tag{31}$$

4.4 The rational correction removes the radial mismatch

The remaining mismatch is between $G\mathbf{v}$ and $Q(\theta_\lambda)h/3375$. The fixed rational row $\xi(x)$ in the appendix satisfies the functional identity

$$G\mathbf{v} - \frac{Q(\theta_\lambda)h}{3375} = \frac{\mathrm{d}}{\mathrm{d}U}\{U\xi(x)\mathbf{v}(\rho x^3)\}. \tag{32}$$

Here is its exact coordinate test. The three weights are

$$(d_0, d_1, d_2) = \frac{1}{3375}(9, 126, 532),$$

and the anchor and parameter action give

$$\sum_{j=0}^{2} d_j(c_j + 6c_{j+1}) = \frac{9c_0 + 180c_1 + 1288c_2 + 3192c_3}{3375}. \tag{33}$$

Pairing this row with $\mathbf{v}$ gives $Q(\theta_\lambda)h/3375$, so the outer weights are exactly the radial load from Section 2. The six rational source coordinates obey

$$\begin{aligned} G - \frac{9c_0 + 180c_1 + 1288c_2 + 3192c_3}{3375} &= \xi - 2(1-x)\xi' \\ &\quad - \frac{6(1-x)}{x}\xi B_6(\rho x^3), \end{aligned} \tag{34}$$

where the prime on ξ means $\mathrm{d}/\mathrm{d}x$. The ninety fixed coefficients of ξ are printed in Table 1 and in the complete exact checker. This checks a specified witness; the proof does not require a search through an ansatz.

To pass from this rational row identity to (32), use $x_U = -2U$, $\lambda_U/\lambda = -6U/x$ and (28). Ordinary differentiation gives

$$\frac{\mathrm{d}}{\mathrm{d}U}\{U\xi(x)\mathbf{v}(\lambda)\} = \left[\xi - 2(1-x)\xi' - \frac{6(1-x)}{x}\xi B_6\right]\mathbf{v}.$$

Combining with (31), define

$$\boxed{\mathcal{E}(U) = \mathfrak{b}(U) + U\xi(1-U^2)\mathbf{v}\big(\rho(1-U^2)^3\big).} \tag{35}$$

Then the actual function satisfies

$$\mathcal{E}'(U) = -\frac{1}{3375}Q(\theta_\lambda)h(\lambda), \qquad \lambda = \rho(1-U^2)^3, \tag{36}$$

off the escape point. The following sections justify its continuation and both endpoint limits.

5 The interior escape singularity is removable

At $x_e = 50/99$, direct substitution gives the two roots λ_e and 1 of f, with $0 < \lambda_e < 1$. Choose a fixed circle of radius $\lambda_e < r_e < 1$. For complex U sufficiently close to U_e, its integrands remain analytic on the circle. The continued integral and the rational correction in (35) are meromorphic in U; their only possible local pole comes from powers of $99x - 50$. For real nearby U this circle is deformable to the balanced circle without crossing either root, so it represents the same function $\mathcal{E}$.

The right-hand side of (36) is holomorphic near U_e. Consequently $\mathcal{E}$ has no negative Laurent coefficient: the derivative of a nonzero term $a_{-k}(U - U_e)^{-k}$ would have a pole of order $k + 1$. Thus $\mathcal{E}$ extends holomorphically through U_e. Both real one-sided limits agree. This conclusion concerns the sum of the two terms in (35); every rational escape term in each coefficient remains included.

6 The upper endpoint and the rational correction

6.1 The beta contour tends to zero at $U = 1$

As $q \to 0$ in a disk, $F(q) = 1 + (5/72)q + O(q^2)$. Formula (14) shows that all three components of $W(q)$ are $O(q^2)$. In the first component the potentially linear terms are $-5q/72 + (5/72)q$ and cancel. The bounds are uniform on a sufficiently small disk.

As $U \to 1^-$, put $r = \sqrt{\lambda} = ux^{3/2}$. On $|s| = r$ both s and $b = \lambda/s$ have modulus r, so the product of the two dual-state bounds is $O(r^4) = O(x^6)$. The matrices L_P, R_P are bounded polynomials, $s - m = O(x)$ and $\ell_0 = O(x)$. Also

$$|f| \geq 2mr - 2\lambda = 2r(m - r) \geq c_1 xr, \qquad |d| \geq c_2 x^4$$

for positive constants c_1, c_2 and sufficiently small $x > 0$. Indeed $m/x \to 54/25$ while $r/x \to 0$. It follows from (18) that $B_s = O(x^{-4}/r)$ uniformly on the circle. Including its length $2\pi r$ and the factor $1/\tau$,

$$\mathfrak{b}(U) = O(x^2) \longrightarrow 0. \tag{37}$$

6.2 All singular and constant terms of the outer correction cancel

The literal matrix C gives

$$C^{-1}(1,0,0,0,0,0)^{\mathsf{T}} = v_0 = (0,0,0,5/72,0,0)^{\mathsf{T}},$$

$$C^{-1}(1,1,1,1,1,1)^{\mathsf{T}} = v_1 = (-1,-1,-1,77/24,77/24,77/24)^{\mathsf{T}}.$$

Since $a_1^2\rho = 9/40000$, the normally convergent series give

$$\mathbf{v}(\rho x^3) = v_0 + \frac{9}{40000}x^3 v_1 + O(x^6). \tag{38}$$

Expand the full rational row as $\xi(x) = \sum_{k=-4}^{\infty} \xi_k x^k$ near zero. Exact substitution of its printed coefficients yields

$$\xi_{-4}v_0 = \xi_{-3}v_0 = \xi_{-2}v_0 = 0, \tag{39}$$

$$\xi_{-1}v_0 = -\frac{105800}{459459}, \quad \frac{9}{40000}\xi_{-4}v_1 = \frac{105800}{459459}, \tag{40}$$

$$\xi_0 v_0 = \frac{8464}{21879}, \quad \frac{9}{40000}\xi_{-3}v_1 = -\frac{8464}{21879}. \tag{41}$$

Both $(99x-50)^{-1}$ and $(99x-50)^{-2}$ contribute to ξ_0. Equations (39)–(41) cancel all negative powers and the constant term of $\xi\mathbf{v}$. Multiplying the $O(x^6)$ remainder in (38) by $\xi = O(x^{-4})$ gives $O(x^2)$. Thus $\xi\mathbf{v} = O(x)$ and

$$\mathcal{E}(1-) = 0. \tag{42}$$

At the other endpoint $x \to 1$, the row ξ is finite and the state is analytic at $\lambda = \rho$. Consequently

$$U\xi(x)\mathbf{v}(\lambda) \longrightarrow 0 \qquad (U \to 0^+). \tag{43}$$

7 The lower endpoint is a full positive residue

Choose once and for all $\rho < r_0 < u$; for example $r_0 = 1/10$. For sufficiently small $U > 0$, both roots of f lie outside $|s| = r_0$, whereas the balanced circle encloses σ_+ alone. On the fixed inner circle $B_s \to 0$ uniformly: it has the explicit factor U, its remaining denominator is nonzero at $U = 0$, and both native arguments stay in compact subsets of the unit disk. Hence its normalized integral tends to zero.

The positively oriented annulus between the two circles contains exactly the pole σ_+. The residue theorem gives their difference as the full residue of the scalar integrand in (22), because $2\pi i/\tau = 1$. Algebraically,

$$\mathrm{Res}_{s=\sigma_+} B_s = \frac{UL_P + \sqrt{M}R_P}{d} \longrightarrow S_* := \frac{4R_P(1)}{d(1)}. \tag{44}$$

No pole lies on either contour in this application. Taking the limit of the actual inside residue gives

$$\mathfrak{b}(0+) = Y(u)^{\mathsf{T}} T(u) S_* T(u)^{\mathsf{T}} Y(u). \tag{45}$$

The polynomial matrices yield the small exact product

$$T(u)S_*T(u)^{\mathsf{T}} = \begin{pmatrix} -1/750 & 1/150 & 98/1125 \\ 1/150 & 98/1125 & 0 \\ 98/1125 & 0 & 0 \end{pmatrix}. \tag{46}$$

Every quantity in this computation is rational, except the subsequently evaluated native solution values. The sign is fixed by the annulus orientation and the selected inside root.

The native Gauss equation and symmetric-square identity proved in the next section imply, at $q = u$,

$$F = g^2, \qquad \theta F = 2g\theta g, \qquad \theta^2 F = 2(\theta g)^2 + 2g\theta^2 g,$$

with $\theta^2 g = (27/196)\theta g + (15/1568)g$ at u. Substitution in (46) gives

$$\mathfrak{b}(0+) = \frac{\{3F(u) + 28\theta F(u)\}^2}{4500}. \tag{47}$$

For an explicit check, the coefficients of $g^4, g^3\theta g, g^2(\theta g)^2$ on both sides are respectively $9/4500, 336/4500, 3136/4500$.

8 A native Gauss certificate for the $1/\pi$ value

The residue square now requires the independent evaluation of $3F(u) + 28\theta F(u)$. We derive it from a normalized Gauss solution; both the signed transformation and its absolute Wronskian scale are proved here.

Define

$$H(p) = \sum_{n\geq 0} \frac{(1/6)_n (5/6)_n}{(n!)^2} p^n. \tag{48}$$

The function F, Euler operator $\theta = q\,\mathrm{d}/\mathrm{d}q$, and normalization $\tau = 2\pi i$ retain their earlier definitions. The consecutive coefficient ratios of H and F tend to one, so both have radius of convergence one and define analytic functions with termwise derivatives on the open unit disk. For real $0 < q < 1$, define

$$p(q) = \frac{1 - \sqrt{1-q}}{2}, \qquad g(q) = H(p(q)), \qquad Z(q) = H(1 - p(q)). \tag{49}$$

Every square root and fractional power of a positive real number below has its positive real value. A prime on H always means its ordinary derivative with respect to its argument.

8.1 The normalized symmetric square

The coefficient recurrence of H gives the actual Gauss equation

$$p(1-p)H''(p) + (1-2p)H'(p) - \frac{5}{36}H(p) = 0, \qquad H(0) = 1. \tag{50}$$

Indeed its Euler form is $\{p\partial_p\}^2 H - p\{p\partial_p + 1/6\}\{p\partial_p + 5/6\}H = 0$, whose coefficient of p^n is $n^2 h_n - (n-5/6)(n-1/6)h_{n-1}$ for $n \geq 1$. The equation is invariant under $p \mapsto 1-p$, so $H(1-p)$ also satisfies it wherever $0 < p < 1$.

Write $s = \sqrt{1-q}$. The identities

$$p(1-p) = q/4, \qquad 1 - 2p = s, \qquad \frac{dp}{dq} = \frac{1}{4s}, \qquad \frac{d^2p}{dq^2} = \frac{1}{8s^3}$$

and the chain rule give, for either $y = g$ or $y = Z$,

$$\theta^2 y = \frac{q}{2(1-q)}\theta y + \frac{5q}{144(1-q)}y. \tag{51}$$

For clarity, the corresponding ordinary derivative equation is

$$y'' = \left(-\frac{1}{q} + \frac{1}{2(1-q)}\right)y' + \frac{5}{144q(1-q)}y,$$

where these primes mean q derivatives.

Define the two differential expressions

$$\mathcal{E}_2 y = 144(1-q)\theta^2 y - 72q\theta y - 5qy,$$
$$\mathcal{E}_3 f = 72(1-q)\theta^3 f - q(108\theta^2 f + 46\theta f + 5f).$$

The product rule, without any transformation formula as an input, gives

$$\mathcal{E}_3(y^2) = y\theta(\mathcal{E}_2 y) + 3(\theta y)\mathcal{E}_2 y. \tag{52}$$

For example, the expansion of either side is

$$144(1-q)y\theta^3 y + 432(1-q)(\theta y)(\theta^2 y) - 216qy\theta^2 y - 216q(\theta y)^2 - 92qy\theta y - 5qy^2.$$

Consequently g^2 is annihilated by $\mathcal{E}_3$. The series F is annihilated by the same expression, since

$$n^3 f_n = (n-5/6)(n-1/2)(n-1/6)f_{n-1} \qquad (n \geq 1). \tag{53}$$

This recurrence uniquely determines every analytic solution from its constant coefficient: the multiplier n^3 is nonzero for $n \geq 1$. The function g is analytic near zero because $p(0) = 0$ and the square root in (49) is analytic there. Since $g(0)^2 = F(0) = 1$, the recurrence proves equality near zero. Both sides are real analytic on the connected interval $(0, 1)$, so

$$F(q) = g(q)^2 \qquad (0 < q < 1). \tag{54}$$

This last continuation can equivalently be obtained by uniqueness for the nonsingular third-order ordinary equation on each compact subinterval of $(0, 1)$.

8.2 A signed algebraic transformation

For every real $h > 8$, set

$$\begin{aligned} v_1(h) &= \frac{(h-8)\sqrt{h+64}}{(h+16)^{3/2}}, \qquad p_1(h) = \frac{1-v_1(h)}{2}, \quad q_1(h) = \frac{1728h}{(h+16)^3}, \\ v_2(h) &= \frac{(h-512)\sqrt{h+64}}{(h+256)^{3/2}}, \quad p_2(h) = \frac{1-v_2(h)}{2}, \quad q_2(h) = \frac{1728h^2}{(h+256)^3}, \\ L(h) &= \left(\frac{h+256}{h+16}\right)^{1/4}. \end{aligned} \tag{55}$$

We will prove the identity of actual functions

$$H(p_2(h)) = L(h)H(p_1(h)) \qquad (h > 8). \tag{56}$$

The numerator $h - 512$ is signed throughout this assertion.

Here is a direct pullback and normalization proof, including the rational identities needed to check its gauge. Put $z = 1/h$ and

$$S = 1 + 64z, \qquad d_1 = 1 + 16z, \qquad d_2 = 1 + 256z, \qquad \Delta = z\frac{d}{dz}.$$

The functions $\widetilde{p}_i(z) = p_i(1/z)$ extend analytically to zero by the explicit formulas

$$\widetilde{p}_1 = \frac{1 - (1-8z)\sqrt{S}/d_1^{3/2}}{2}, \qquad \widetilde{p}_2 = \frac{1 - (1-512z)\sqrt{S}/d_2^{3/2}}{2}. \tag{57}$$

The elementary polynomial identities

$$d_1^3 - (1-8z)^2 S = 1728z^2, \qquad d_2^3 - (1-512z)^2 S = 1728z \tag{58}$$

imply

$$\widetilde{p}_1(1-\widetilde{p}_1)=\frac{432z^2}{d_1^3},\qquad \widetilde{p}_2(1-\widetilde{p}_2)=\frac{432z}{d_2^3}.$$

In particular $0<\widetilde{p}_i<1$ when $0<z<1/8$; the first one is also less than $1/2$ because $1-8z>0$.

Differentiating the literal signed expressions gives

$$\frac{d\widetilde{p}_1}{dz}=\frac{864z}{\sqrt{S}\,d_1^{5/2}},\qquad \frac{d\widetilde{p}_2}{dz}=\frac{432}{\sqrt{S}\,d_2^{5/2}}. \tag{59}$$

For an explicit check of these derivative numerators,

$$\begin{aligned}-8Sd_1+(1-8z)(32d_1-24S)&=-1728z,\\ -512Sd_2+(1-512z)(32d_2-384S)&=-864.\end{aligned}$$

Thus there is no extra factor $1-8z$ in either derivative. Their logarithmic derivatives on $0<z<1/8$ are

$$\frac{\widetilde{p}_1''}{\widetilde{p}_1'}=\frac{1}{z}-\frac{32}{S}-\frac{40}{d_1},\qquad \frac{\widetilde{p}_2''}{\widetilde{p}_2'}=-\frac{32}{S}-\frac{640}{d_2}, \tag{60}$$

where the derivatives in this display are with respect to z.

For any such pullback $Y(z)=H(\widetilde{p}(z))$, the chain rule in (50) yields

$$\Delta^2Y+B\Delta Y+CY=0,$$

with the explicit coefficients

$$B=\frac{z\widetilde{p}'(1-2\widetilde{p})}{\widetilde{p}(1-\widetilde{p})}-1-\frac{z\widetilde{p}''}{\widetilde{p}'},\qquad C=-\frac{5}{36}\frac{z^2(\widetilde{p}')^2}{\widetilde{p}(1-\widetilde{p})}. \tag{61}$$

All divisions here are legitimate by (58) and (59). Substitution gives

$$\begin{aligned}B_1&=z\left(-\frac{8}{d_1}+\frac{32}{S}\right),\quad C_1=-\frac{240z^2}{Sd_1^2},\\ B_2&=z\left(-\frac{128}{d_2}+\frac{32}{S}\right),\quad C_2=-\frac{60z}{Sd_2^2}.\end{aligned} \tag{62}$$

For instance the first fractions in the expressions for B_1,B_2 in (61) are respectively $2(1-8z)/d_1$ and $(1-512z)/d_2$; together with (60) these produce exactly the displayed B_i.

Let

$$\mathcal{L}(z)=\left(\frac{d_2}{d_1}\right)^{1/4},\qquad \mu(z)=\frac{\mathcal{L}'(z)}{\mathcal{L}(z)}=\frac{60}{d_1d_2}.$$

Its ordinary z derivative is

$$\mu'=-\frac{60(16d_2+256d_1)}{d_1^2d_2^2}=-\frac{60(272+8192z)}{d_1^2d_2^2}.$$

The gauge identities are

$$B_1=B_2+2z\mu,\qquad C_1=C_2+z^2(\mu'+\mu^2)+z(1+B_2)\mu. \tag{63}$$

The first follows from $-8d_2+128d_1=120$. To verify the second without an implicit computational certificate, multiply it by $Sd_1^2d_2^2$ and use this polynomial identity:

$$\begin{aligned}-240z^2d_2^2+60zd_1^2=Sz^2(3600-960d_2-15360d_1)+60zSd_1d_2\\ -7680z^2Sd_1+1920z^2d_1d_2.\end{aligned} \tag{64}$$

It is an identity after $S = 1 + 64z$, $d_1 = 1 + 16z$ and $d_2 = 1 + 256z$ are inserted.

We also give the ordinary h equations explicitly. Since $\Delta = -h\, d/dh$ on a function pulled back by $z = 1/h$, the corrected coefficients (62) imply

$$Y_i'' + a_i Y_i' + b_i Y_i = 0, \qquad Y_i(h) = H(p_i(h)), \tag{65}$$

where these primes are h derivatives and

$$\begin{aligned} a_1 &= \frac{1}{h}\left(1 + \frac{8}{h+16} - \frac{32}{h+64}\right), \quad & b_1 &= -\frac{240}{h(h+64)(h+16)^2}, \\ a_2 &= \frac{1}{h}\left(1 + \frac{128}{h+256} - \frac{32}{h+64}\right), \quad & b_2 &= -\frac{60}{(h+64)(h+256)^2}. \end{aligned} \tag{66}$$

Indeed $a_i = (1 - B_i(1/h))/h$ and $b_i = C_i(1/h)/h^2$. Writing

$$\mu_h(h) = \frac{L'(h)}{L(h)} = -\frac{60}{(h+16)(h+256)}, \qquad \mu_h'(h) = \frac{60(2h+272)}{(h+16)^2(h+256)^2}, \tag{67}$$

the same identities become

$$a_1 = a_2 + 2\mu_h, \qquad b_1 = b_2 + \mu_h' + \mu_h^2 + a_2\mu_h. \tag{68}$$

For the second conversion, explicitly $\mu_h = -\mu(1/h)/h^2$ and

$$\mu_h' + \mu_h^2 + a_2\mu_h = \frac{\mu'(1/h) + \mu(1/h)^2}{h^4} + \frac{(1 + B_2(1/h))\mu(1/h)}{h^3} = \frac{C_1(1/h) - C_2(1/h)}{h^2}.$$

Thus the product rule proves that LY_1 satisfies the second ordinary equation:

$$(LY_1)'' + a_2(LY_1)' + b_2 LY_1 = L\{Y_1'' + (2\mu_h + a_2)Y_1' + (\mu_h' + \mu_h^2 + a_2\mu_h + b_2)Y_1\} = 0.$$

It remains to fix the actual solution, rather than only its equation. Both $H(\widetilde{p}_2(z))$ and $\mathcal{L}(z)H(\widetilde{p}_1(z))$ are analytic near zero, take the value one there, and satisfy $\Delta^2 Y + B_2\Delta Y + C_2 Y = 0$; the displayed formulas give $\widetilde{p}_1(0) = \widetilde{p}_2(0) = 0$ and $\mathcal{L}(0) = 1$. The rational functions B_2, C_2 are analytic and vanish at zero. If their two solutions differed, the first nonzero term of their analytic difference would be cz^m with $m \geq 1$. The coefficient of z^m in its equation would then be $m^2 c$, because the other two terms have higher order. This contradicts $c \neq 0$. Hence the solutions agree near $z = 0$. On $h > 8$ the coefficients (66) are continuous and the equation is nonsingular. Ordinary initial-value uniqueness continues the equality from sufficiently large h to the entire connected interval $h > 8$, proving (56). This uniqueness step requires only the usual local uniqueness for a linear first-order system: on a compact interval its bounded coefficient matrix gives uniqueness by the integral equation and Gronwall's inequality.

All functions in (55) are smooth at $h = 512$. In particular $p_2(512) = 1/2$ and H is analytic there; the proof makes no change of branch at that point. Differentiating the actual identity now gives the fully explicit ordinary derivative identity

$$-\frac{432h\, H'(p_2(h))}{\sqrt{h+64}(h+256)^{5/2}} = L(h)\left\{-\frac{60H(p_1(h))}{(h+16)(h+256)} - \frac{864H'(p_1(h))}{\sqrt{h+64}(h+16)^{5/2}}\right\}. \tag{69}$$

Here the two chain derivatives are exactly $p_1'(h) = -864/[\sqrt{h+64}(h+16)^{5/2}]$ and $p_2'(h) = -432h/[\sqrt{h+64}(h+256)^{5/2}]$.

8.3 The Euler endpoint and the normalized Wronskian

We derive the normalization from an integral, including its differentiated endpoint. For $x, y > 0$ write $B(x,y) = \int_0^1 t^{x-1}(1-t)^{y-1}\,dt$. The change of variable $t = r/(1+r)$ and then $r = w^6$ gives

$$B(5/6, 1/6) = \int_0^\infty \frac{r^{-1/6}}{1+r}\,dr = 6\int_0^\infty \frac{w^4}{1+w^6}\,dw.$$

Here is an elementary residue evaluation of the last integral. The upper half-plane poles of $w^4/(1+w^6)$ are $\exp(i\pi/6), i, \exp(5i\pi/6)$, and the residue at a pole ζ is $1/(6\zeta)$. Their sum is

$$\frac{1}{6}\{e^{-i\pi/6} + e^{-i\pi/2} + e^{-5i\pi/6}\} = -\frac{i}{3}.$$

For a semicircle of radius $R \geq 2$ the absolute value of the integrand is at most $2/R^2$, so its arc integral tends to zero. The residue theorem therefore gives $\int_{-\infty}^{\infty} w^4/(1+w^6)\,dw = 2\pi/3$. The integrand is even, and consequently

$$B(5/6, 1/6) = 2\pi. \tag{70}$$

For $0 \leq x < 1$, binomial expansion in the positive integral yields

$$H(x) = \frac{1}{2\pi}\int_0^1 t^{-1/6}(1-t)^{-5/6}(1-xt)^{-1/6}\,dt. \tag{71}$$

To see the coefficients without assuming an Euler formula, integration by parts gives $B(a+1,b) = aB(a,b)/(a+b)$ for $a, b > 0$. Indeed the integral of $(t^a(1-t)^b)'$ is zero, and $B(a,b) = B(a+1,b) + B(a,b+1)$. It follows that $B(5/6+n, 1/6)/B(5/6,1/6) = (5/6)_n/n!$. Inserting the binomial coefficients $(1/6)_n/n!$ in the integral therefore produces exactly (48); monotone convergence justifies the interchange for $0 \leq x < 1$.

On every compact interval $0 \leq x \leq r < 1$, differentiation under this integral is justified by the integrable majorant

$$\frac{(1-r)^{-7/6}}{12\pi} t^{5/6}(1-t)^{-5/6}.$$

Thus the derivative of the actual function is

$$H'(x) = \frac{1}{12\pi}\int_0^1 t^{5/6}(1-t)^{-5/6}(1-xt)^{-7/6}\,dt. \tag{72}$$

Take $x = 1-\delta$ with $0 < \delta \leq 1/2$, first set $r_0 = 1-t$ and then $r_0 = \delta v$. This gives the exact equation

$$\delta H'(1-\delta) = \frac{1}{12\pi}\int_0^{1/\delta} (1-\delta v)^{5/6} v^{-5/6}\{1+(1-\delta)v\}^{-7/6}\,dv. \tag{73}$$

Extend this integrand by zero for $v > 1/\delta$. It is bounded by $v^{-5/6}(1+v/2)^{-7/6}$, which is integrable: it is $O(v^{-5/6})$ at zero and $O(v^{-2})$ at infinity. Dominated convergence and the substitution $w = v/(1+v)$ now give

$$\int_0^\infty v^{-5/6}(1+v)^{-7/6}\,dv = \int_0^1 w^{-5/6}\,dw = 6.$$

Consequently

$$\lim_{\delta \downarrow 0} \delta H'(1-\delta) = \frac{1}{2\pi}. \tag{74}$$

The undifferentiated integral has a sufficient elementary bound of its own: $1-(1-\delta)t \geq \delta$ for $0 \leq t \leq 1$. By (70) and (71),

$$0 < H(1-\delta) \leq \delta^{-1/6}, \qquad 0 < \delta H(1-\delta) \leq \delta^{5/6} \longrightarrow 0 \qquad (\delta \downarrow 0). \tag{75}$$

For $0 < q < 1$ define the Euler Wronskian

$$\mathcal{W}(q) = g(q)\theta Z(q) - (\theta g(q))Z(q).$$

Equation (51) gives

$$\theta\mathcal{W} = \frac{q}{2(1-q)}\mathcal{W}, \qquad \frac{d}{dq}\{\sqrt{1-q}\,\mathcal{W}(q)\} = 0.$$

Its constant is fixed by (74): as $q \downarrow 0$, $p(q) = q/4 + O(q^2)$ and

$$\frac{q\,p'(q)}{p(q)} = \frac{1+\sqrt{1-q}}{2\sqrt{1-q}} \longrightarrow 1, \qquad \theta Z(q) = -q\,p'(q)H'(1-p(q)) \longrightarrow -\frac{1}{2\pi}.$$

Also $g(q) \to 1$ and $\theta g(q) = q\,p'(q)H'(p(q)) = O(p(q))$: H' is bounded near zero and $q/p(q) \to 4$. The independent bound (75) therefore gives $(\theta g(q))Z(q) = O(p(q)^{5/6}) \to 0$. We have proved

$$g(q)\theta Z(q) - (\theta g(q))Z(q) = -\frac{1}{2\pi\sqrt{1-q}}. \tag{76}$$

In particular the derivative normalization comes directly from the differentiated integral and its integrable bound, not from differentiating a remainder in a bare logarithmic expansion.

8.4 The actual contact at $h = 64$

Put $u = 27/125$ and $s_u = \sqrt{1-u} = 7\sqrt{10}/25$. Substitution in (55) gives

$$\begin{aligned} &q_1(64) = q_2(64) = u, \quad && L(64) = \sqrt{2}, \\ &v_1(64) = s_u, && v_2(64) = -s_u, \\ &p_1(64) = p(u), && p_2(64) = 1 - p(u). \end{aligned} \tag{77}$$

Therefore the actual transformation already proves

$$Z(u) = \sqrt{2}\,g(u). \tag{78}$$

On a neighborhood of $h = 64$ contained in $(8, 512)$ the positive-root definition in (49) satisfies $p_1(h) = p(q_1(h))$ and $p_2(h) = 1 - p(q_2(h))$. This follows from $4p_i(1-p_i) = q_i$, $v_1 > 0$ and $v_2 < 0$, which are also consequences of (58). We may consequently differentiate the actual equality $Z(q_2(h)) = L(h)g(q_1(h))$ on that neighborhood. The three logarithmic derivatives are

$$\frac{q_1'}{q_1} = \frac{1}{h} - \frac{3}{h+16}, \qquad \frac{q_2'}{q_2} = \frac{2}{h} - \frac{3}{h+256}, \qquad \frac{L'}{L} = -\frac{60}{(h+16)(h+256)},$$

so at $h = 64$ their values are $-7/320$, $7/320$ and $-3/1280$. The differentiated identity is thus

$$\frac{7}{320}\theta Z(u) = \sqrt{2}\left\{-\frac{3}{1280}g(u) - \frac{7}{320}\theta g(u)\right\},$$

or, equivalently,

$$\theta Z(u) = -\sqrt{2}\,\theta g(u) - \frac{3\sqrt{2}}{28}g(u). \tag{79}$$

This relation uses the ordinary derivative of an identity on an open interval, with all moving parameters and the derivative of L retained.

Finally insert (78) and (79) into (76), and use $F = g^2$ and $\theta F = 2g\theta g$ from (54). It follows that

$$-\sqrt{2}\left\{\theta F(u) + \frac{3}{28}F(u)\right\} = -\frac{1}{2\pi s_u}.$$

Since $s_u = 7\sqrt{10}/25$, this is the required native certificate

$$\boxed{3F\left(\frac{27}{125}\right) + 28\theta F\left(\frac{27}{125}\right) = \frac{5\sqrt{5}}{\pi}.} \tag{80}$$

9 Completion of the series evaluation

Equation (80) substituted in (47) gives

$$\mathfrak{b}(0+) = \frac{(5\sqrt{5}/\pi)^2}{4500} = \frac{1}{36\pi^2}.$$

Together with (43), this gives $\mathcal{E}(0+) = 1/(36\pi^2)$. The upper limit is (42), and the escape singularity has been removed. On every closed subinterval of $(0,1)$ the extended primitive satisfies (36). The loaded radial integrand is continuous and bounded on $[0,1]$, by its uniformly convergent polynomial-geometric series. Applying the fundamental theorem of calculus first away from the endpoints and then taking their limits yields

$$\int_0^1 Q(\theta_\lambda) h(\rho(1-U^2)^3)\,\mathrm{d}U = 3375\{\mathcal{E}(0+) - \mathcal{E}(1-)\} = \frac{3375}{36\pi^2} = \frac{375}{4\pi^2}.$$

Equivalently, before inserting the Gauss value, the same endpoint calculation gives

$$R = \frac{3375}{4500}\{3F(u) + 28\theta F(u)\}^2 = \frac{3}{4}\{3F(u) + 28\theta F(u)\}^2,$$

which is the central reduction (2). Finally (10) identifies this entire integral with the literal factorial series in Theorem 1.1, proving the theorem.

A A finite rational certificate for the source row

This appendix supplies the rational identities used in the contour comparison. In particular, it includes an actual primitive, rather than a rank computation or a numerical fit. Everything in the certificate is defined below over rational function fields. The contour's analytic boundary and normalization are the statements proved in the main text; finite algebra by itself would not establish those statements.

Write

$$\rho = \frac{729}{15625}, \qquad x = 1 - U^2, \qquad \lambda = \rho x^3, \qquad b = \frac{\lambda}{s}, \qquad \tau = 2\pi i.$$

All coefficient rows act on solution columns. Thus their covariant differential has a *plus* sign. Tensor coordinates are ordered $(00, 01, 02, 10, 11, 12, 20, 21, 22)$.

A.1 The complete source coefficient

The ordinary elliptic symmetric-square state is $(y^2, 2yy', y'^2)^{\mathsf{T}}$; its third entry is not the second derivative of its first entry. Its connection and Euler gauge are

$$\mathsf{A}(q) = \begin{pmatrix} 0 & 1 & 0 \\ 2c(q) & -a(q) & 2 \\ 0 & c(q) & -2a(q) \end{pmatrix}, \quad a(q) = \frac{1 - 3q/2}{q(1-q)}, \quad c(q) = \frac{5}{144q(1-q)},$$

$$\mathsf{G}(q) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & q & 0 \\ \dfrac{5q}{72(1-q)} & \dfrac{q^2}{2(1-q)} & 2q^2 \end{pmatrix}, \qquad N(q) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ \dfrac{5q}{72(1-q)} & \dfrac{23q}{36(1-q)} & \dfrac{3q}{2(1-q)} \end{pmatrix}.$$

They satisfy $\mathsf{G}' + \mathsf{G}\mathsf{A} = (N/q)\mathsf{G}$. The solution-state form and its useful product with the gauge are

$$J_C(q) = \begin{pmatrix} -5q/36 & -q/2 & 1-q \\ -q/2 & q-1 & 0 \\ 1-q & 0 & 0 \end{pmatrix}.$$

$$T(q) = J_C(q)\mathsf{G}(q) = \begin{pmatrix} -5q/72 & 0 & 2q^2(1-q) \\ -q/2 & q(q-1) & 0 \\ 1-q & 0 & 0 \end{pmatrix}.$$

Here are the coefficient polynomials of the literal rational source. Put

$$\begin{aligned} d &= x^4(25-9x)(99x-50)^2, & K_2 &= 81x^2 - 198x + 80, \\ K_3 &= 243x^3 - 1053x^2 + 1266x - 400, & K_6 &= 59049x^6 - 511758x^5 + 1724085x^4 \\ & & & \quad - 2860596x^3 + 2441988x^2 - 1011200x + 160000, \\ k &= \frac{1953125}{209952}, & j &= \frac{244140625}{944784}, \\ v &= \frac{3906250}{6561}, & c_* &= \frac{30517578125}{8503056}. \end{aligned}$$

Then

$$L_P = \begin{pmatrix} 0 & 0 & -k(9x-25)K_2K_3 \\ 0 & 0 & j(9x-25)K_2 \\ k(9x-25)K_2K_3 & -j(9x-25)K_2 & 0 \end{pmatrix}.$$

$$R_P = \begin{pmatrix} 0 & 0 & kK_6 \\ 0 & vx(99x-50) & -jK_3 \\ kK_6 & -jK_3 & c_* \end{pmatrix}.$$

For compact notation put $L = UL_P/d$ and $R = R_P/d$. Define

$$m = \frac{27x(80-99x+27x^2)}{1000}, \quad \ell_0 = \frac{27x(9x-16)}{1000}, \quad \ell = U\ell_0, \quad M = 25-9x,$$

$$z = s - m, \qquad f = z^2 - \ell^2 M = s^2 - 2ms + \lambda.$$

The entire rational coefficient one-form is

$$\beta_N = \frac{L\,\mathrm{d}f + R(2\ell M\,\mathrm{d}z - 2zM\,\mathrm{d}\ell - z\ell\,\mathrm{d}M)}{f} = T_U\,\mathrm{d}U + UT_s\,\mathrm{d}s, \tag{81}$$

where, with primes denoting ∂_x at fixed s,

$$T_s = \frac{2L_P(s-m) + 2\ell_0 M R_P}{df}, \tag{82}$$

$$T_U = \frac{-2(1-x)L_P f_x - 2R_P\{-2\ell_0(1-x)Mm' + (s-m)[(\ell_0 - 2(1-x)\ell_0')M + 9(1-x)\ell_0]\}}{df}. \tag{83}$$

Thus the correction has not been omitted or absorbed into an unspecified boundary term. The relation to the main coefficients is $B_s = UT_s$ and $B_U = T_U$. Taking their covariant derivatives gives

$$\begin{aligned} \mathsf{K} = {} & -T_s + 2(1-x)\partial_x T_s - \frac{6(1-x)b}{x} T_s \mathsf{A}(b)^{\mathsf{T}} \\ & + \partial_s T_U - \mathsf{A}(s)T_U + \frac{b}{s} T_U \mathsf{A}(b)^{\mathsf{T}}, \end{aligned} \tag{84}$$

$$\mathfrak{q} = \mathrm{flatten}\big(T(s)\mathsf{K}T(b)^{\mathsf{T}}\big). \tag{85}$$

In particular, the source row against the native product Euler state is $\mathfrak{q}\,\mathrm{d}U \wedge \mathrm{d}s$. Thus $\mathfrak{q} = \mathrm{flatten}(\mathsf{S})$ in the notation of the main text. Formula (84) is simply $-\nabla_U B_s + \nabla_s B_U$, with $x_U = -2U$ and $b_U = -6Ub/x$; these identities fix both its sign and its chain-rule terms.

A.2 Angular normal form and its six-state action

Let

$$J = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \qquad \mathbf{w} = \begin{pmatrix} 5/72 & 23/36 & 3/2 \end{pmatrix}, \qquad E = \mathbf{e}_2^{\mathsf{T}}\mathbf{w},$$

Here $\mathbf{e}_2 = (0, 0, 1)$ is the standard row, as in the main text. Over $\mathbb{Q}(\lambda)(s)$,

$$\mathcal{M}_s = \frac{N(s) \otimes I - I \otimes N(\lambda/s)}{s} = \frac{R_0}{s} + \frac{R_1}{s-1} + \frac{R_\lambda}{s-\lambda},$$

$$R_0 = J \otimes I - I \otimes (J - E), \qquad R_1 = -E \otimes I, \qquad R_\lambda = -I \otimes E.$$

For a six-row $v = (a_0, a_1, a_2, b_0, b_1, b_2) = (a, b_*)$, define

$$\mathrm{NF}(v) = \frac{\mathbf{w} \otimes a}{s-1} + \frac{b_* \otimes \mathbf{w}}{s-\lambda}. \tag{86}$$

The six standard coordinate rows under this formula are the six angular basis representatives. Their induced D_λ action is

$$B_6(\lambda) = \begin{pmatrix} J + \dfrac{\lambda E}{1-\lambda} & -\dfrac{\lambda E}{1-\lambda} \\ -\dfrac{E}{1-\lambda} & J + \dfrac{\lambda E}{1-\lambda} \end{pmatrix}. \tag{87}$$

This assertion includes an explicit angular primitive. For a constant row $v = (a, b_*)$, put $p_v = -s(b_* \otimes \mathbf{w})/(s-\lambda)$. Entrywise,

$$\lambda \partial_\lambda \mathrm{NF}(v) + \mathrm{NF}(v)(I \otimes N(\lambda/s)) - \mathrm{NF}(vB_6) = \partial_s p_v + p_v \mathcal{M}_s. \tag{88}$$

For the first three basis rows $p_v = 0$.

The coefficient one-form for the anchor is $\omega_0 = e_{00}\,\mathrm{d}s/s$, whose normalized contour integral is the scalar function h. If $\eta_0 = e_{00}R_0^{-1}$, its exact angular identity is $e_{00}/s = \mathrm{NF}(c_0) + \eta_0 \mathcal{M}_s$. The coordinate rows of $\omega_0, D_\lambda\omega_0, \ldots, D_\lambda^5\omega_0$ form the following full matrix:

$$C = \begin{pmatrix} 49456/125 & 153648/125 & 113472/125 & 72/5 & -3312/25 & 113472/125 \\ -1576/25 & -4608/25 & -3312/25 & 0 & 72/5 & -3312/25 \\ 46/5 & 108/5 & 72/5 & 0 & 0 & 72/5 \\ -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 \end{pmatrix}. \tag{89}$$

Writing its rows as $c_0, \ldots, c_5$, one has $c_j B_6 = c_{j+1}$ for $0 \le j < 5$ and $\det C = 373248/125$. These are identities in the full six-state system.

For completeness, the angular elimination used below is finite and unambiguous. At a finite puncture p, a leading pole row $v/(s-p)^m$, $m \ge 2$, is removed by subtracting the covariant derivative of

$$\frac{v(R_p - (m-1)I)^{-1}}{(s-p)^{m-1}}.$$

At infinity a polynomial leading row vs^d is removed by subtracting the derivative of $v((d+1)I + R_0 + R_1 + R_\lambda)^{-1}s^{d+1}$. The eigenvalues of R_0 are $1/6, 1/2, 5/6$, each repeated three times; those of R_1, R_λ are 0 and $-3/2$; those of $R_0 + R_1 + R_\lambda$ are $-1/6, -1/2, -5/6$, each repeated three times. Every inverse just used therefore exists. A final constant primitive removes the simple zero residue because R_0 is invertible. Finite primitives decay at infinity and create only simple poles at other finite punctures; polynomial primitives also create only simple finite poles.

Thus each step strictly lowers the relevant higher pole or polynomial degree and the procedure terminates. The same leading-coefficient argument proves uniqueness of the remaining form $v_1/(s-1) + v_\lambda/(s-\lambda)$, whose residues are in general arbitrary rows of length nine. In the uniqueness argument a constant primitive must also vanish, because it would create a zero residue through invertible R_0. Thus this general normal form has eighteen coordinates; its reduction to the two three-coordinate tensor rows below is a separate property of the complete source.

Applying this stated procedure to (85) gives a rational row η_S and six-row $G(x)$ satisfying

$$\mathfrak{q} = \mathrm{NF}(G) + \partial_s \eta_S + \eta_S \mathcal{M}_s. \tag{90}$$

The listing below constructs and retains the primitive; its variable `eta`, divided by the coefficient-field scaling `scale` $= x^5(9x-25)(99x-50)^3$, is η_S. It verifies this entire nine-entry identity and the factorization of both residues in (86). No unspecified angular witness is needed.

A.3 The outer primitive and all its coefficients

Use the six-coordinate solution column $\mathcal{V}$ obtained by multiplying $\mathbf{v}(\rho x^3)$ by $U = \sqrt{1-x}$. Thus $\mathcal{V} = U\mathbf{v}(\rho x^3)$, and its ordinary x-connection is

$$M_x = \frac{3B_6(\rho x^3)}{x} - \frac{I}{2(1-x)}.$$

Since $\mathrm{d}U = -\mathrm{d}x/(2U)$, the source row and three normalized period rows in this frame are

$$r_S = -\frac{G}{2(1-x)}, \qquad H_j = -\frac{c_j + 6c_{j+1}}{2(1-x)}, \qquad 0 \le j \le 2.$$

Here D_λ denotes the coefficient action corresponding to θ_λ; the full factor $1 + 6D_\lambda$ is retained. Define

$$(d_0, d_1, d_2) = \left(\frac{1}{375}, \frac{14}{375}, \frac{532}{3375}\right).$$

An actual rational primitive is

$$\begin{aligned} \xi(x) &= \sum_{k=1}^{15} A_k \phi_k(x), \\ (\phi_1, \dots, \phi_6) &= (x^{-1}, x^{-2}, x^{-3}, x^{-4}, (99x-50)^{-1}, (99x-50)^{-2}), \\ (\phi_7, \dots, \phi_{15}) &= (1, x, \dots, x^8). \end{aligned} \tag{91}$$

All ninety rational entries of the rows A_k are printed in Table 1 and repeated literally in `COEFFICIENTS` in the listing. No coefficient is a variable to be solved for. With that witness the exact identity is

$$r_S - d_0 H_0 - d_1 H_1 - d_2 H_2 = \xi' + \xi M_x. \tag{92}$$

It can be checked by clearing denominators in six rational functions. Neither existence of a primitive nor a claim about a computed rank is a premise of this equality.

The angular primitive enters the two-form identity with its wedge sign: the exact one-form $-\eta_S\,\mathrm{d}U$, paired with the product solution column, contributes $(\partial_s\eta_S + \eta_S\mathcal{M}_s)\mathrm{d}U \wedge \mathrm{d}s$. The outer term is the ordinary derivative of $\xi\mathcal{V}$. Hence (90) and (92) are the actual primitives required for the comparison. Their poles at $99x - 50 = 0$ remain present; the contour truncation and endpoint argument in the main text justify applying the identities to the selected literal contour.

A.4 The fixed ninety-coefficient witness

The row $A_k = (A_{k0}, \dots, A_{k5})$ multiplies the basis function ϕ_k in (91). Signs and basis order agree with the executable certificate. The entries are exact rational numbers.

k	ϕ_k	A_{k0}	A_{k1}	A_{k2}	A_{k3}	A_{k4}	A_{k5}
1	x^{-1}	$\frac{105800}{459459}$	$\frac{867560}{459459}$	$\frac{103229200}{459459}$	$-\frac{169280}{51051}$	$-\frac{1388096}{51051}$	$-\frac{2268352}{51051}$
2	x^{-2}	0	0	$-\frac{472485400}{459459}$	0	0	0
3	x^{-3}	0	0	$\frac{338560000}{196911}$	0	0	0
4	x^{-4}	0	0	$-\frac{4232000000}{4135131}$	0	0	0
5	$(99x-50)^{-1}$	$-\frac{78890805485931260}{886402053948411}$	$-\frac{1227125311357021190}{886402053948411}$	$-\frac{70427029255928891500}{6204814377638877}$	$-\frac{607873567055897705440}{74327471429736107583}$	$-\frac{298973396640778498112}{8258607936637345287}$	$-\frac{145365088004028281168}{635277533587488099}$
6	$(99x-50)^{-2}$	$-\frac{231954922267076000}{229807939912551}$	$-\frac{331597299956488000}{76602646637517}$	$-\frac{108998157631733600}{5892511279809}$	$-\frac{39913346954892419000}{917623104070816143}$	$\frac{28252939480465998800}{101958122674535127}$	$-\frac{14427468858797595200}{5997536627913831}$
7	1	$-\frac{44128649359258592}{31024071888194385}$	$\frac{623049674458568524}{31024071888194385}$	$\frac{6660655461825680912}{31024071888194385}$	$\frac{57598947351747925330}{10618210204248015369}$	$\frac{370356988491186478316}{8258607936637345287}$	$\frac{422172179336846381488}{5899005669026675205}$
8	x	$\frac{7410960991604618219}{7834361587927875000}$	$-\frac{2244024947258679397}{230422399644937500}$	$-\frac{8368444588355333518}{75330399883921875}$	$-\frac{1674067760127596905471}{469239087308940073125}$	$-\frac{7571431721389433993041}{260688381838300040625}$	$-\frac{60217890176084166521579}{1303441909191500203125}$
9	x^2	$-\frac{1014783336398633}{79134965534625000}$	$\frac{99347762374288343}{39567482767312500}$	$\frac{226386615454772638}{9891870691828125}$	$\frac{1399948922915894608}{4739788760696364375}$	$\frac{6901571092120745041}{2633215978164646875}$	$\frac{59461141531221275693}{13166079890823234375}$
10	x^3	$-\frac{1051211136}{19184234069}$	$-\frac{440067361152}{479605851725}$	$-\frac{18358156683264}{2398029258625}$	$\frac{10318818710613893 12}{1196916353711203125}$	$\frac{258130738760714016}{40711440602421875}$	$\frac{20750847913747165488}{1994860589518671875}$
11	x^4	$\frac{1029377673}{34880425580}$	$\frac{430927242561}{872010639500}$	$\frac{4494215281788}{1090013299375}$	$-\frac{250759381828805733}{558002962103125000}$	$-\frac{14957478836244716238}{4533774067087890625}$	$-\frac{24572653010256580956}{4533774067087890625}$
12	x^5	$-\frac{16224867}{3170947780}$	$-\frac{6792198219}{79273694500}$	$-\frac{70837018452}{99092118125}$	$\frac{300193629643629927}{3297290230609375000}$	$\frac{1380547922050429344}{2060806394130859375}$	$\frac{2261266906791518928}{2060806394130859375}$
13	x^6	0	0	0	$-\frac{223303109482368}{7493841433203125}$	$-\frac{8431993935111168}{37469207166015625}$	$-\frac{13467359214383616}{37469207166015625}$
14	x^7	0	0	0	$\frac{218665144746549}{13625166242187500}$	$\frac{2064215740879206}{17031457802734375}$	$\frac{3296911156760547}{17031457802734375}$
15	x^8	0	0	0	$-\frac{3446560950471}{1238651476562500}$	$-\frac{32535799768674}{1548314345703125}$	$-\frac{51965324712513}{1548314345703125}$

Table 1: All coefficients of the fixed rational row $\xi(x)$.

A.5 A complete exact verification recipe

The following Python listing uses only exact SymPy arithmetic. It is a self-contained transcription of the finite identities: it reads no previous certificate or saved output, executes no solver, and contains the complete coefficient witness. Equality of rational functions is tested by exact polynomial cancellation. Its source is obtained from (81), not inserted as a fitted six-row. During angular reduction, multiplying by $x^5(9x-25)(99x-50)^3$ is only a coefficient-field scaling and is undone in the result. For generic x and $\lambda \neq 0, 1$, all steps are defined; the resulting rational identities consequently hold throughout their domains by clearing denominators.

```
"""Standalone exact transcription check for the printed rational appendix.

This file supplies its own literal inputs and never reads or executes an old
certificate, solver, or output.  No period, target fit, or rank is an input.
The printed 90 coefficients constitute a witness, not unknowns to be solved.
"""
import sympy as S
from sympy.polys.fields import field
import time

started = time.perf_counter()
X, Z, LAM = S.symbols('x s lambda')
rho = S.Rational(729, 15625)
I = S.eye(3)
J = S.Matrix([[0, 1, 0], [0, 0, 1], [0, 0, 0]])
w = S.Matrix([[S.Rational(5,72), S.Rational(23,36), S.Rational(3,2)]])
E = S.zeros(3)
E[2,:] = w
R0 = S.kronecker_product(J,I) - S.kronecker_product(I,J-E)
R1 = -S.kronecker_product(E,I)
Rl = -S.kronecker_product(I,E)
Rt = R0 + R1 + Rl
B6 = (J + LAM*E/(1-LAM)).row_join(-LAM*E/(1-LAM)).col_join(
    (-E/(1-LAM)).row_join(J + LAM*E/(1-LAM)))
C = S.Matrix([
    [S.Rational(49456,125),S.Rational(153648,125),S.Rational(113472,125),
     S.Rational(72,5),-S.Rational(3312,25),S.Rational(113472,125)],
    [-S.Rational(1576,25),-S.Rational(4608,25),-S.Rational(3312,25),
     0,S.Rational(72,5),-S.Rational(3312,25)],
    [S.Rational(46,5),S.Rational(108,5),S.Rational(72,5),0,0,S.Rational(72,5)],
    [-1,0,0,0,0,0],[0,-1,0,0,0,0],[0,0,-1,0,0,0]])

def cancel_matrix(A):
    return A.applyfunc(S.cancel)

def nf(v):
    return (S.kronecker_product(w,v[:,:3])/(Z-1)
            + S.kronecker_product(v[:,3:],w)/(Z-LAM))

N = lambda q: J + q*E/(1-q)
gauge = S.Matrix([[1,0,0],[0,LAM,0],
                  [5*LAM/(72*(1-LAM)),LAM**2/(2*(1-LAM)),2*LAM**2]])
pq,cq = (1-S.Rational(3,2)*LAM)/(LAM*(1-LAM)),5/(144*LAM*(1-LAM))
ao = S.Matrix([[0,1,0],[2*cq,-pq,2],[0,cq,-2*pq]])
jc = S.Matrix([[-5*LAM/36,-LAM/2,1-LAM],[-LAM/2,LAM-1,0],[1-LAM,0,0]])
tq = S.Matrix([[-5*LAM/72,0,2*LAM**2*(1-LAM)],
               [-LAM/2,LAM*(LAM-1),0],[1-LAM,0,0]])
assert cancel_matrix(gauge.diff(LAM)+gauge*ao-N(LAM)*gauge/LAM) == S.zeros(3)
assert cancel_matrix(jc*gauge-tq) == S.zeros(3)
Ms = S.kronecker_product(N(Z),I)/Z - S.kronecker_product(I,N(LAM/Z))/Z
Ml = S.kronecker_product(I,N(LAM/Z))
assert cancel_matrix(Ms-R0/Z-R1/(Z-1)-Rl/(Z-LAM)) == S.zeros(9)
for j in range(6):
    v = S.eye(6)[j,:]
    p = S.zeros(1,9) if j < 3 else -Z*S.kronecker_product(v[:,3:],w)/(Z-LAM)
    assert cancel_matrix(LAM*nf(v).diff(LAM)+nf(v)*Ml-nf(v*B6)
                         -p.diff(Z)-p*Ms) == S.zeros(1,9)
e00 = S.eye(9)[0,:]
anchor = e00*R0.inv()
assert cancel_matrix(e00/Z-nf(C[0,:])-anchor*Ms) == S.zeros(1,9)
for j in range(5):
```

```
    assert cancel_matrix(C[j,:]*B6-C[j+1,:]) == S.zeros(1,6)
assert C.det() == S.Rational(373248,125)

# The rational beta source, with beta_N = T_U dU + U*T_s ds.
x, s, lam = X, Z, rho*X**3
m = 27*x*(80-99*x+27*x**2)/1000
a = 27*x*(9*x-16)/1000
M = 25-9*x
D = x**4*M*(99*x-50)**2
f = s**2-2*m*s+lam
a2 = 81*x**2-198*x+80
a3 = 243*x**3-1053*x**2+1266*x-400
a6 = (59049*x**6-511758*x**5+1724085*x**4-2860596*x**3
      +2441988*x**2-1011200*x+160000)
k,jj,vv,cc = map(S.Rational, ['1953125/209952','244140625/944784',
                              '3906250/6561','30517578125/8503056'])
LP = S.Matrix([[0,0,-k*(9*x-25)*a2*a3],
               [0,0,jj*(9*x-25)*a2],
               [k*(9*x-25)*a2*a3,-jj*(9*x-25)*a2,0]])
RP = S.Matrix([[0,0,k*a6],[0,vv*x*(99*x-50),-jj*a3],
               [k*a6,-jj*a3,cc]])

# These temporary matrices are the displayed beta numerators.
# Their common scalar denominator is D*f.
Ts = cancel_matrix(2*LP*(s-m)+2*a*M*RP)
TU = cancel_matrix((-2*(1-x)*LP*S.diff(f,x)-2*RP*(
     -2*a*(1-x)*M*S.diff(m,x)+(s-m)*(
     (a-2*(1-x)*S.diff(a,x))*M+9*(1-x)*a))))

# The beta numerators are polynomials in s over Q(x).
# Clear the known connection denominator, divide by f^2 exactly, and only
# then create rational rows.  This preserves all nine source entries.
F, x = field('x',S.QQ)
K, s = field('s',F.to_domain())
z = K.zero
lam = rho*x**3
lift = K.ground_new
parse = lambda q: K.from_expr(q)
constant = lambda A: [[parse(A[i,j]) for j in range(A.cols)] for i in range(A.rows)]
ts_fraction,tu_fraction = constant(Ts),constant(TU)
assert all(q.denom.degree() == 0 for row in ts_fraction+tu_fraction for q in row)
ts = [[q.numer.mul_ground(F.one/q.denom.LC) for q in row] for row in ts_fraction]
tu = [[q.numer.mul_ground(F.one/q.denom.LC) for q in row] for row in tu_fraction]
P = K.ring
sp = s.numer
pp = P.one
zz = P.zero
u = 1-x
mfield = F.from_expr(m)
dlogD = F.from_expr(S.diff(D,X)/D)
fpoly = sp**2-2*mfield*sp+lam

def dx_polynomial(p):
    return P.from_dict({e:c.diff(x) for e,c in p.items()})

fx,fs = dx_polynomial(fpoly),fpoly.diff(sp)
angular_den = sp**2*(1-sp)*(sp-lam)
asbar = [[zz,sp*(1-sp),zz],
         [S.Rational(5,72)*pp,-pp+S.Rational(3,2)*sp,2*sp*(1-sp)],
         [zz,S.Rational(5,144)*pp,-2*pp+3*sp]]
cbbar = [[zz,lam*(sp-lam),zz],
         [S.Rational(5,72)*sp**2,-sp**2+S.Rational(3,2)*lam*sp,2*lam*(sp-lam)],
         [zz,S.Rational(5,144)*sp**2,-2*sp**2+3*lam*sp]]
scale = lift(x**5*(9*x-25)*(99*x-50)**3)
scale_over_D = -x*(99*x-50)
alpha = [[z]*3 for _ in range(3)]
for i in range(3):
    for j in range(3):
        scalar_part = (-ts[i][j]*fpoly
            +2*u*(dx_polynomial(ts[i][j])*fpoly-ts[i][j]*fx
                  -dlogD*ts[i][j]*fpoly)
            +tu[i][j].diff(sp)*fpoly-tu[i][j]*fs)
        numerator = (angular_den*scalar_part
```

```
            -sp*(sp-lam)*sum((asbar[i][k]*tu[k][j] for k in range(3)),zz)*fpoly
            +(1-sp)*sum((tu[i][k]*cbbar[j][k] for k in range(3)),zz)*fpoly
            -(6*u/x)*sp*(1-sp)*sum((ts[i][k]*cbbar[j][k] for k in range(3)),zz)*fpoly)
        quotient,remainder = numerator.div(fpoly**2)
        assert remainder == zz
        alpha[i][j] = K.new(quotient.mul_ground(scale_over_D),angular_den)
print('All nine source entries pass exact zero-remainder f^2 division.', flush=True)

def metric_gauge(q):
    return [[-5*q/72,z,2*q*q*(1-q)],
            [-q/2,q*(q-1),z],[1-q,z,z]]

b = lift(lam)/s
left,right = metric_gauge(s),metric_gauge(b)
Q = [[sum((left[i][a]*alpha[a][c]*right[j][c]
           for a in range(3) for c in range(3)),z)
      for j in range(3)] for i in range(3)]
print('Source transcribed from the complete beta formulas.', flush=True)

r0,r1,rl = map(constant,[R0,R1,Rl])
ms = [[r0[i][j]/s+r1[i][j]/(s-1)+rl[i][j]/(s-lift(lam))
       for j in range(9)] for i in range(9)]

def row_times(v,A):
    return [sum((v[i]*A[i][j] for i in range(len(v))),z)
            for j in range(len(A[0]))]

def covariant(v):
    return [v[j].diff(s)+q for j,q in enumerate(row_times(v,ms))]

def evaluate(q,a):
    def ev(p):
        return sum((c*(a**e[0] if e[0] else F.one) for e,c in p.items()),F.zero)
    return lift(ev(q.numer)/ev(q.denom))

def pole_order(q,a):
    p,linear,n = q.denom,(s-lift(a)).numer,0
    while q:
        quotient,remainder = p.div(linear)
        if remainder:
            break
        p,n = quotient,n+1
    return n

q = [Q[i][j] for i in range(3) for j in range(3)]
r, eta = q[:],[z]*9

def subtract(p):
    global r,eta
    exact = covariant(p)
    r = [r[j]-exact[j] for j in range(9)]
    eta = [eta[j]+p[j] for j in range(9)]

for a,residue in [(F.one,R1),(lam,Rl),(F.zero,R0)]:
    while True:
        order = max(pole_order(t,a) for t in r)
        if order < 2:
            break
        lead = [evaluate(t*(s-lift(a))**order,a) for t in r]
        inv = constant((residue-(order-1)*S.eye(9)).inv())
        subtract([t/(s-lift(a))**(order-1) for t in row_times(lead,inv)])
while True:
    polynomials = [t.numer.div(t.denom)[0] for t in r]
    degree = max(p.degree(0) if p else -1 for p in polynomials)
    if degree < 0:
        break
    lead = [lift(p.get((degree,),F.zero)) for p in polynomials]
    inv = constant(((degree+1)*S.eye(9)+Rt).inv())
    subtract([t*s**(degree+1) for t in row_times(lead,inv)])
subtract(row_times([evaluate(t*s,F.zero) for t in r],constant(R0.inv())))
u = [evaluate(t*(s-1),F.one) for t in r]
v = [evaluate(t*(s-lift(lam)),lam) for t in r]
normal = [u[j]/(s-1)+v[j]/(s-lift(lam)) for j in range(9)]
```

```
assert all(not (r[j]-normal[j]) for j in range(9))
exact = covariant(eta)
assert all(not (q[j]-normal[j]-exact[j]) for j in range(9))
g = [u[6+j]/parse(w[0,2])/scale for j in range(3)]
g += [v[3*j+2]/parse(w[0,2])/scale for j in range(3)]
assert all(not (u[3*i+j]-parse(w[0,i])*g[j]*scale)
           and not (v[3*i+j]-g[3+i]*parse(w[0,j])*scale)
           for i in range(3) for j in range(3))
print('Angular source identity established with its rational primitive.', flush=True)

# Literal 15 by 6 coefficient table: basis order is specified below.
COEFFICIENTS = '''
105800/459459 867560/459459 103229200/459459 -169280/51051 -1388096/51051 -2268352/51051
0 0 -472485400/459459 0 0 0
0 0 338560000/196911 0 0 0
0 0 -4232000000/4135131 0 0 0
-78890805485931260/886402053948411 -1227125311357021190/886402053948411
     -7042702925592889150 0/6204814377638877 -607873567055897705440/7432747142973610758 3
     -2989733966407784981 12/8258607936637345287 -145365088004028281168/635277533587488099
-231954922267076000/229807939912551 -331597299956488000/76602646637517 -108998157631733600/5892511279809
     -3991334695489241900 0/917623104070816143 282529394804659988 00/101958122674535127
     -1442746858797595200/5997536627913831
-44128649359258592/31024071888194385 623049674458568524/31024071888194385
     66606554618256809 12/31024071888194385 575989473517479253 30/10618210204248015369
     37035698849118647831 6/82586079366373452 87 42217217933684638148 8/5899005669026675205
741096099160461821 9/783436158792787500 0 -22440249472586793 97/230422399644937500
     -8368445588355333518/75330399883921875 -167406776012759690 5471/469239087308940073125
     -7571431721389433993041/260688381838300040625 -602178901760841665215 79/130344190919150020 3125
-1014783336398633/79134965534625000 9934776237428834 3/39567482767312500 22638661545477263 8/9891870691828125
      1399948922915894608/4739788760696364375 690157109212074504 1/2633215978164646875
     5946114153122127569 3/13166079890823234375
-1051211136/19184234069 -440067361152/479605851725 -18358156683264/2398029258625
     10318818710613893 12/1196916353711203125 258130738760714016/40711440602421875
     20750847913747165488/1994860589518671875
1029377673/34880425580 430927242561/872010639500 4494215281788/1090013299375
     -250759381828805733/558002962103125000 -14957478836244716238/4533774067087890625
     -2457265301025658095 6/4533774067087890625
-16224867/3170947780 -6792198219/79273694500 -70837018452/99092118125
     30019362964362992 7/3297290230609375000 13805479220504293 44/2060806394130859375
     22612669067915189 28/2060806394130859375
0 0 0 -223303109482368/7493841433203125 -8431993935111168/37469207166015625
     -1346735921438361 6/37469207166015625
0 0 0 218665144746549/13625166242187500 2064215740879206/17031457802734375
     3296911156760547/17031457802734375
0 0 0 -3446560950471/1238651476562500 -32535799768674/1548314345703125 -51965324712513/1548314345703125
'''
coefficients = [[F.from_expr(S.Rational(t)) for t in line.split()]
                for line in COEFFICIENTS.strip().splitlines()]
assert len(coefficients) == 15 and all(len(row) == 6 for row in coefficients)
basis = [x**(-p) for p in range(1,5)]
basis += [(99*x-50)**(-p) for p in range(1,3)] + [x**p for p in range(9)]
xi = [sum((coefficients[k][j]*basis[k] for k in range(15)),F.zero)
      for j in range(6)]
mx = [[3*F.from_expr(B6[i,j].subs(LAM,lam.as_expr()))/x
       -(F.one/(2*(1-x)) if i == j else F.zero)
       for j in range(6)] for i in range(6)]
d = list(map(S.Rational,['1/375','14/375','532/3375']))
h = [[-F.from_expr(C[k,j]+6*C[k+1,j])/(2*(1-x))
      for j in range(6)] for k in range(3)]
for j in range(6):
    residual = (-g[j].as_expr()/(2*(1-X))
                -sum(d[k]*h[k][j].as_expr() for k in range(3))
                -xi[j].diff(x).as_expr()
                -sum((xi[i]*mx[i][j]).as_expr() for i in range(6)))
    assert S.cancel(residual) == 0
assert [-81*t for t in d] == list(map(S.Rational,['-27/125','-378/125','-1596/125']))
print('All finite printed identities hold over the rational function fields.')
print('Elapsed seconds:', round(time.perf_counter()-started,3))
```

B Provenance and review boundary

The finite rational inputs were transcribed from the frozen Row12 proof package associated with author commit `a17703b4e69a5d879d71ebd37fcaba32af45be0c` and strict-audit package `7b71f2c0f1c72798dda3e8759da4465b489366c0`. In that package the relevant inputs are the source-lift proof and `s03/LIFT_CHECK.output.json`, the six-coordinate proof `s08/PROOF.md`, and the complete outer coefficients `s14/OUTER_ROW_CHECK.output.json`. The finite formulas and verification procedure used in this document are printed in Appendix A, so those historical files are provenance references rather than additional compilation inputs.

The analytic scalar route was organized in `SCALAR_INTEGRAL_ROUTE.md` and checked together with the literal coefficient and moment bridge in the additional scalar review performed 7 October 2026. The signed Gauss calculation uses the corrected actual derivatives recorded in the completed signed-Gauss Lean support package; the earlier unchecked draft is not a premise of this proof. The 7 October document-level review and its affected-cone corrections belong to the preserved original manuscript package `C5/row12_tex_2026-10-07/DOCUMENT_REVIEW.md`. That review found no substantive defect in the retained scalar proof. The mathematical tables and the complete literal checker are contained in this TeX file, so compiling or reading it requires none of those historical project files.

The evidential status remains [ARG] for the analytic derivation and [ONE] for exact arithmetic and document-transcription checks. The scalar proof status is Complete with no open mathematical premise. Human acceptance remains Pending. The connected Lean formalization was completed on 7 October 2026. Its unconditional declarations are `Row12.row12_hasSum`, `Row12.row12_summable` and `Row12.row12_tsum`. The final statement and axiom outputs, all five verification checks and the fresh full-result review are retained in the separate `C5/row12_lean_2026-10-06` package. That formal result covers exactly the scalar equality proved here.